Federated Learning with AE-WGAN for Anomaly-Based Network Intrusion Detection

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Abstract

This project implements and evaluates an Anomaly-based Intrusion Detection System combining a Denoising Autoencoder with a Wasserstein GAN (AE-WGAN) for synthetic attack generation, extended with Federated Learning (FedAvg) for privacy-preserving training. Through an iterative debugging process spanning six implementation versions, we identified critical discrepancies between the reference paper and its published code, uncovering that dataset balancing—not just synthetic data quality—is the key driver of rare attack detection. On NSL-KDD, our balanced AE-WGAN improves R2L detection by +28.5% F1 (RF), within 8.5 points of the paper's reported +37%. Our Federated approach achieves within 0.2–0.4% of centralized performance while preserving data privacy. We also validate on CICIDS-2017, where our balanced 70K CNN multiclass F1 (0.685) closely matches the paper's reported 0.65, confirming the dataset balancing hypothesis across both datasets.

1 Introduction

Network intrusion detection faces three fundamental challenges. First, **class imbalance**: in NSL-KDD, Normal traffic has 67,343 training samples while U2R attacks have only 52—a ratio exceeding 1,200:1. Classifiers trained on such data tend to ignore rare attacks. Second, **high dimensionality**: NSL-KDD has 41 raw features and CICIDS-2017 has 80, many irrelevant or redundant. Third, **privacy constraints**: organizations cannot share raw network data due to regulations like GDPR, necessitating collaborative training without data sharing.

This project addresses these challenges by combining three techniques: (1) AE-WGAN for generating realistic synthetic attacks to balance training data, (2) ANOVA feature selection to reduce dimensionality, and (3) Federated Learning (FedAvg) for privacy-preserving distributed training. The project is based on Arafah et al. (2025) [1] for the AE-WGAN architecture and Buyuktanir et al. (2025) [2] for the Federated Learning framework. The FL integration is our added contribution.

A significant part of this work involved reproducing the paper's results, which required identifying and resolving multiple discrepancies between the paper's description and its published code. This process yielded insights into the critical role of preprocessing and dataset balancing that are documented throughout this report.

2 Related Work

Zhang et al. [4] proposed CWGAN-CSSAE, achieving 87.18% F1 on NSL-KDD. Cui et al. [5] introduced GMM-WGAN-IDS with 86.88% F1. Lee and Park [6] combined AE with conditional GANs for CICIDS-2017. The AE-WGAN approach by Arafah et al. [1] reports 94% F1 on NSL-KDD binary classification and 98% on CICIDS-2017, with a notable +37% F1 improvement on R2L rare attacks.

In Federated Learning for IDS, Buyuktanir et al. [2] survey FedAvg, FedProx, and FedYogi algorithms. Our work bridges these directions by integrating FedAvg with AE-WGAN.

3 Problem Definition and Algorithm

3.1 Task

Input: Network traffic features $\mathbf{x} \in R^d$ ($d = 24$ for NSL-KDD, $d = 10$ for CICIDS-2017 after ANOVA selection).

Output: Binary classification (Normal/Attack) or multiclass classification (Normal, DoS, Probe, R2L, U2R for NSL-KDD; 15 classes for CICIDS-2017).

Objective: Improve classification of rare attacks via AE-WGAN data augmentation, while enabling privacy-preserving training through Federated Learning.

3.2 Preprocessing

The pipeline follows four steps: (1) remove constant and quasi-constant features, (2) ordinal encoding of categorical features, (3) min-max scaling to $[0, 1]$, (4) ANOVA feature selection. The exact 24 features for NSL-KDD were taken from Table 5 of the paper, including the `difficulty_level` feature ("Level"). For CICIDS-2017, 10 features were selected per Table 5.

3.3 AE-WGAN Architecture

The architecture follows the authors' GitHub implementation [3], which differs significantly from the paper's prose description (see Section 4.4). It implements a BiGAN-style model:

Encoder: Flatten($4 \times 6 \times 1$) $\rightarrow$ Dense(20, relu) $\rightarrow$ Dense(32, linear). Encodes real data into a 32-dimensional latent vector.

Generator: Dense(20) $\rightarrow$ Dense(40) $\rightarrow$ Dense(80) $\rightarrow$ Dense(120) $\rightarrow$ Dense(160) $\rightarrow$ Dense(24, relu) $\rightarrow$ Reshape$(4, 6, 1)$. For CICIDS-2017: Reshape$(2, 5, 1)$ with linear activations.

Discriminator: Receives concatenated $[\mathbf{z}, \text{Flatten}(\mathbf{img})]$ (dim 56) $\rightarrow$ Dense(80, relu) $\rightarrow$ Dense(40, relu) $\rightarrow$ Dense(1, sigmoid).

Training uses Wasserstein loss $\mathcal{L} = E[y \cdot \hat{y}]$, Adamax optimizer (lr $= 9 \times 10^{-5}$), batch size 32, 101 epochs, 1:1 update ratio.

3.4 Dataset Balancing Strategy

A critical finding of this project: the paper states that "the total number of samples used was 70,343, which balanced the number of samples in each attack category." This implies **downsampling** majority classes (Normal: 67,343 $\rightarrow$ 14,069; DoS: 45,927 $\rightarrow$ 14,069) while **augmenting** minority classes (R2L: 995 $\rightarrow$ 14,069; U2R: 52 $\rightarrow$ 14,069) via AE-WGAN. This produces a perfectly balanced dataset of $\sim$70K samples across 5 categories.

3.5 Federated Learning (FedAvg)

We simulate FedAvg across 4 non-IID clients, each specializing in one attack category plus a share of Normal traffic. For K clients with dataset sizes n_k:

$$\mathbf{w}_{\text{global}} = \sum_{k=1}^{K} \frac{n_k}{\sum_j n_j} \cdot \mathbf{w}_k$$

We run 5 communication rounds with 20 local epochs per round ($\approx$ 101 total).

4 Experimental Evaluation

4.1 Data

NSL-KDD: 125,973 training / 22,544 testing samples. 5 categories: Normal (67,343), DoS (45,927), Probe (11,656), R2L (995), U2R (52). 24 ANOVA features.

CICIDS-2017: 2,827,876 samples (80/20 split). 15 classes. 10 ANOVA features.

4.2 Methodology

Six classifiers following the paper's Table 4: DT, RF, CNN, GRU, LSTM, RNN (relu for NSL-KDD, tanh for CICIDS-2017). Four experimental conditions: (1) Baseline, (2) AE-WGAN with 10% threshold, (3) AE-WGAN with balanced 70K dataset, (4) Federated AE-WGAN.

4.3 Iterative Debugging Process

Reaching the final results required six successive implementation versions, each revealing a different source of error.

Table 1: Evolution across implementation versions (best binary F1)

V.	Configuration	Best F1	Key Issue Found
1	Our WGAN-GP architecture	0.791	Wrong architecture
2	Corrected (deeper G, Adamax)	0.805	Still not matching GitHub code
3	GitHub BiGAN + N+20K	0.792	Dataset inflated to 531K
4	GitHub + 10% threshold	0.800	Wrong ANOVA features
5	GitHub + Table 5 exact features	**0.912**	Correct preprocessing
6	Table 5 + balanced 70K	0.906	Best rare attack detection

V1–V2: Architecture mismatch. The paper describes a standard WGAN, but the GitHub code implements a BiGAN with concatenated (z, image) inputs to the Discriminator, a 5-layer Generator, and 1:1 training ratio. Discrimination test accuracy was 100% in V1 (synthetic data trivially distinguishable).

V3: Oversampling ratio. The GitHub code generates $N + 20{,}000$ samples per type including Normal, inflating the dataset to 531K. This *degraded* results. The paper specifies 10% of the majority class.

V4: Feature selection. Dynamic ANOVA selected 4–6 different features than Table 5. Critically, the paper keeps `difficulty_level` ("Level"), `land`, and `wrong_fragment`—features we had removed. Hardcoding Table 5 features raised baseline F1 from 79% to 91%.

V5–V6: Dataset balancing. The paper's mention of 70,343 total samples with "balanced categories" led to the hypothesis that majority classes are downsampled. This proved to be the single most impactful change for rare attack detection.

4.4 Results — NSL-KDD Centralized (Table 5 Features)

Table 2: NSL-KDD Binary Classification — Baseline vs AE-WGAN (10% threshold)

Classifier	Base F1	AE-WGAN F1	Δ F1	Paper F1
DT	0.911	0.912	+0.1%	0.94
RF	0.827	0.866	**+3.9%**	0.93
CNN	0.900	0.909	**+0.9%**	0.89
GRU	0.862	0.876	**+1.4%**	0.89

Four out of six classifiers improve in binary. CNN (0.909) exceeds the paper's reported value of 0.89. GRU (0.876) approaches but does not reach it.

4.5 Results — R2L Rare Attack Detection

Table 3 shows the progressive improvement of R2L detection across our optimization strategies.

Table 3: R2L F1-score evolution across strategies

Strategy	DT	RF	CNN	GRU	Paper CNN
Baseline (no augmentation)	0.320	0.080	0.285	0.350	0.28
AE-WGAN per subtype	0.413	0.152	0.171	0.000	—
AE-WGAN per category	0.391	0.204	0.306	0.052	—
AE-WGAN balanced 70K	**0.374**	**0.365**	**0.404**	**0.427**	**0.65**
Best Δ vs baseline	+5.4%	**+28.5%**	+11.9%	+7.7%	+37%

The balanced 70K approach yields the best R2L results: RF improves by +28.5% (from 0.080 to 0.365), within **8.5 points** of the paper's +37%. All four classifiers improve on R2L with this strategy.

4.6 Results — Per-Category Comparison with Paper (CNN, Balanced 70K)

Table 4: Per-category F1 comparison: Paper vs Our best (CNN, balanced 70K)

Category	Paper	Ours	Gap
Normal	0.890	0.886	−0.4 pts
DoS	0.890	**0.898**	**+0.8 pts**
Probe	0.680	**0.769**	**+8.9 pts**
R2L	0.650	0.404	−24.6 pts
U2R	0.000	**0.122**	**+12.2 pts**

We match or exceed the paper on 4 out of 5 categories. Only R2L shows a remaining gap in absolute F1, though the improvement trend (+28.5% RF) approaches the paper's +37%.

4.7 Results — Federated Learning

Table 5: NSL-KDD Binary — Baseline vs Federated AE-WGAN vs Centralized

Clf	Base F1	FL F1	Δ F1	Central F1
DT	0.911	0.895	−1.6%	0.912
RF	0.827	0.862	**+3.6%**	0.866
CNN	0.893	0.907	**+1.4%**	0.909
GRU	0.895	0.889	−0.6%	0.876

The key finding: **FL achieves near-centralized performance while preserving privacy**. RF: FL 0.862 vs centralized 0.866 (gap of 0.4%). CNN: FL 0.907 vs centralized 0.909 (gap of 0.2%). This demonstrates that privacy-preserving IDS training is practical without significant performance loss under non-IID data distribution.

4.8 Results — CICIDS-2017

Table 6: CICIDS-2017: Baseline (full dataset) vs Balanced 70K vs Paper

Task	Clf	Base F1	Bal 70K F1	Paper F1
Binary	DT	0.449	0.590	0.84
	RF	0.973	0.835	0.88
	CNN	0.954	0.591	0.98
	GRU	0.965	0.682	0.98
Multiclass	DT	0.976	0.936	0.52
	RF	0.962	0.899	0.63
	CNN	0.982	**0.685**	**0.65**
	GRU	0.984	0.178	0.69

The CICIDS-2017 results reveal that the paper likely uses different dataset configurations for binary and multiclass classification. Our CNN multiclass on the balanced 70K dataset (0.685) closely matches the paper's reported value (0.65), confirming that the paper uses a balanced dataset for multiclass evaluation. However, the paper's binary scores (0.98) are only achievable with the full dataset, not the balanced one. Our baseline on the full dataset (0.954–0.973 for DL classifiers) is consistent with the paper's reported 98%.

4.9 Discussion

Finding 1: Preprocessing dominates model performance. Switching from dynamic ANOVA to the paper's exact Table 5 features improved baseline F1 by +12 percentage points (79% → 91%). The `difficulty_level` feature proved extremely informative. This demonstrates that feature engineering matters more than generative model sophistication.

Finding 2: Dataset balancing is the key to rare attack detection. The single most impactful change was downsampling Normal/DoS to balance with augmented R2L/U2R. With the standard 10% threshold, R2L still faced a 10:1 imbalance against Normal. With balanced 70K, the ratio becomes 1:1, and RF R2L F1 jumped from 0.080 to 0.365 (+28.5%).

Finding 3: Paper-code discrepancies are significant. Three major differences were found between the paper description and GitHub code: (1) BiGAN architecture vs standard WGAN, (2) 5-layer Generator vs 3-layer, (3) 1:1 training ratio vs 5:1. Additionally, the code

generates $N + 20{,}000$ samples per type while the paper describes 10% threshold. These discrepancies highlight reproducibility challenges in ML research.

Finding 4: FL is viable for IDS. Federated AE-WGAN achieves within 0.2–0.4% of centralized performance on RF and CNN, demonstrating that privacy-preserving training does not require significant performance sacrifice under non-IID conditions.

Finding 5: Per-category vs per-subtype training. Training AE-WGAN on R2L as a whole (995 samples) rather than 13 individual subtypes (20–50 each) improved CNN R2L from -11.9% to $+2.8\%$, showing that pooling minority samples is beneficial when individual subtypes are too small.

Finding 6: The paper uses different configurations per dataset and task. On CICIDS-2017, the paper reports 98% binary F1 and 65% multiclass F1. Our experiments show that 98% is achievable with the full dataset (2.3M samples), while 65% matches our balanced 70K dataset result (CNN: 0.685). This strongly suggests the paper uses the full dataset for binary classification but a balanced dataset for multiclass, without explicitly stating this distinction.

Remaining limitations. The R2L absolute F1 (0.404) remains below the paper's 0.65, likely due to preprocessing differences we could not resolve without the authors' exact code. U2R detection remains near zero for most classifiers due to only 52 training samples. LSTM showed instability with augmented data (-12.3% in binary), likely due to sensitivity to synthetic data noise. On CICIDS-2017, the AE-WGAN augmentation with linear activations did not improve DL classifiers on the full dataset, suggesting that when baseline performance is already near-optimal (>96%), synthetic data introduces more noise than signal.

5 Conclusions

This project successfully reproduced and extended the AE-WGAN architecture for intrusion detection. Key contributions:

1. **Reproduction with critical analysis:** Through six iterative versions, we identified that preprocessing (feature selection) accounts for +12% F1, dataset balancing accounts for +28.5% R2L improvement, and the code-paper discrepancies are substantial.
2. **Best results:** AE-WGAN with balanced 70K dataset improves R2L by +28.5% (RF), within 8.5 points of the paper's +37%. We exceed the paper on Probe (+8.9%), DoS (+0.8%), and U2R (+12.2%).
3. **Federated Learning:** FL achieves within 0.2% of centralized performance, demonstrating practical privacy-preserving IDS.
4. **Cross-dataset validation:** On CICIDS-2017, our balanced 70K CNN multiclass (0.685) matches the paper's 0.65, while our full-dataset baseline (0.954–0.973) matches the paper's 0.98 binary. This reveals that the paper uses different dataset configurations for different evaluation tasks.

Future work could explore: (1) FedProx for better non-IID handling, (2) differential privacy for formal guarantees, (3) conditional generators that receive attack subtype labels, and (4) few-shot learning for extremely rare attacks like U2R.

References

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